

Pranav Karra

973-723-0071 | pkarra@andrew.cmu.edu | linkedin.com/in/pranavkarra | github.com/Pranav-Karra | pkarra.dev

EDUCATION

Carnegie Mellon University

B.S. in Computer Science, Concentration in Machine Learning

Pittsburgh, PA

May 2028

Relevant Coursework: Deep Learning Systems, Foundations in Machine Learning, NLP, Computer Systems, Probability Theory, Discrete Math, Functional Programming, Vector Calculus, Data Structures, Linear Algebra

EXPERIENCE

Anduril Industries

Software Engineering Intern

Costa Mesa, CA

May 2026 – Aug. 2026

- Shipped **3** end-to-end features across a **Rust/C++/gRPC/Protocol Buffers** stack for a real-time 3D simulation platform in **Unreal Engine**, owning API design, engine implementation, testing, and code review
- Trained a CUT/CycleGAN sim-to-real image-translation pipeline in **PyTorch** (ResNet/PatchGAN) on a **122k+** image dataset via Flyte GPU workflows on NVIDIA A100 GPUs, reducing **FID from 76 to 37**
- Recovered **60%** of the sim-to-real gap, improving mAP@0.5 from **69% → 84%** toward a 94% real-image baseline
- Deployed as a real-time on-GPU inference pass in Unreal (ONNX → NNE/RDG), achieving **26ms/38 FPS** at 1024×1024 using **GAN Compression**

Carnegie Mellon University

15-150 Functional Programming Teaching Assistant

Pittsburgh, PA

Aug 2025 – May 2026

- Taught proof-heavy core concepts in **Standard ML (SML)** across 2 semesters for **300+ students** each
- Organized recitations covering higher-order functions, continuation-passing style, and the ML module system
- Graded **3 exams/semester** and 8 programming tasks, assessing correctness proofs and algorithmic implementations

Try Catch Robotics

Robotics Intern

Bridgewater, NJ

May 2025 – Aug. 2025

- Developed 30+ embedded software examples using **Arduino** and **C++** for instructional code on robotics projects
- Engineered over **20 robotics kits** with micro-controllers, sensors, and electrical components for **40 students**
- Utilized **Git** for version control and code management to ensure consistency and reproducibility across projects

PROJECTS

swiftER | Python, FastAPI, Typescript, PostgreSQL

Jan. 2026 – Feb. 2026

- Architected a distributed, real-time coordination system in **Python (FastAPI)** to auto-match ambulances with best-fit hospitals to augment manual radio dispatch; 3rd in Healthcare Track at **NexHacks (1500+ hackers)**
- Built an **NLP-driven Reasoning Agent** that validates Wispr Flow voice-transcribed symptoms against a **Pydantic** schema, cutting hospital-assignment **latency from 7s to 3s (57%)** across **200 simulated patient intakes**
- Designed a hospital-matching microservice combining **OSRM** route optimization with a weighted scoring model over live ETA, bed capacity, specialist availability, and patient acuity across **12 hospitals**
- Synchronized **concurrent** state across **3+ services** using **Supabase Realtime** Postgres Changes to push live ambulance/hospital updates to dashboards, with polling as a fail-safe fallback for capacity broadcasts

Phantom | Python, FastAPI, OpenClaw, Gemini Live API

Mar. 2026 – Apr. 2026

- Engineered a low-latency multimodal inference pipeline using **FastAPI**, **asyncio**, and **WebSockets** that processed live Meta Ray-Ban smart glasses streams to generate executable actions with **<700ms end-to-end latency**
- Designed a **confidence-policy engine** decoupling LLM reasoning from tool execution, rerouting **66% of actions** into autonomous execution or lightweight confirmation, validated across **80 test interactions**
- Extended the **OpenClaw** runtime with a custom **skill** that intercepted every tool invocation, enabling policy-based integrations across **6+ tools** (web search, messaging, etc) into the glasses' action pipeline

TECHNICAL SKILLS

Languages: Python, C/C++, Rust, Java, Assembly, Typescript, JavaScript, SQL, HTML/CSS, SML, Bash

Tools/Frameworks: Pytorch, FastAPI, gRPC, Docker, Linux, Nix, AWS, PostgreSQL, MongoDB, Git, React, Django

Interests: Large Language Models, Agentic AI, Computer Vision, Model Inference